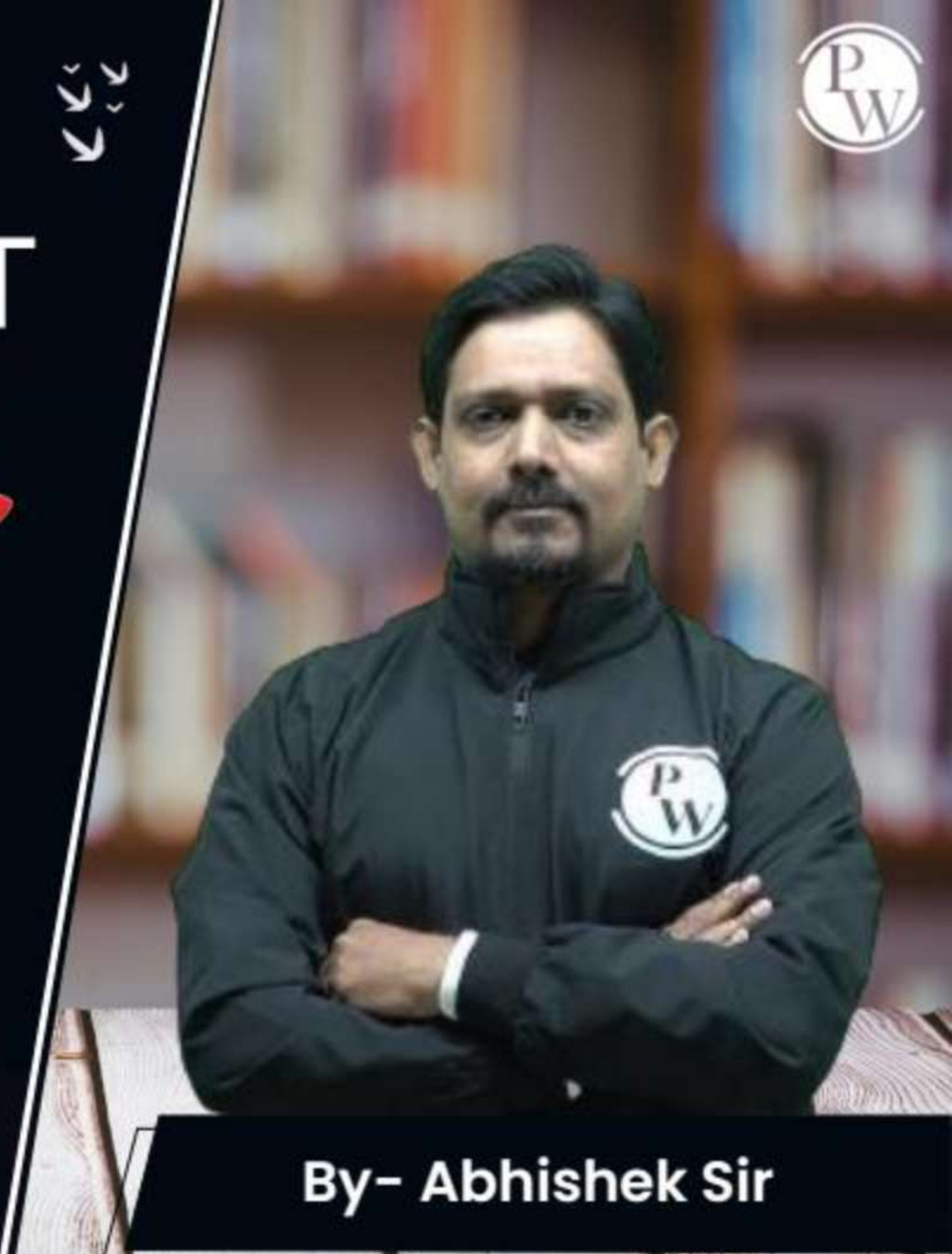
Computer Science & IT



C Programming

Array & Pointers

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Extern Register

pointer

problem

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

pointer arithmetic ← Array

Array

Slide



Question

Consider the following C program?

```
#include <stdio.h>
int a = 100;
void fun(int *x ){
     *x = *x+10;
     a = a-20;
}
int main(){
     a = a+50;
     fun(&a);
     printf("%d",a );
     return 0;
}
```

The value printed by above program is

((A) 120

(B) 140

(C) 130

(D) 140



Question

Consider the following C program?

```
#include <stdio.h>
int a = 100;
void fun(int *x, int y ){
      *x = *x+10;
      y = y-20;
      a= a+20;
}
int main(){
      fun(&a,a);
      printf("%d",a);
      return 0;
}
```

The value printed by above program is

((A) 120

(B) 140

(C) 130

(D) 140

#Q Consider the C program shown below.

```
#include <stdio.h>
int x;
void Q(int z) {
          z += x;
           printf("%d ", z);
}
void P(int *y) {
     int x = *y+2;
     Q(x);
     *y = x-1;
      printf("%d ", x);
     }
main(void) {
      x = 5;
      P(&x)
       printf("%d ", x);
}
```

The output of this program is

A 12 7 6

B 22 12 11

C 14 6 6

D 7 6 6



Question

Consider the following C program?

```
#include <stdio.h>
void func(int *x,int *y,int *z){
    *y=*y+4;
    *z=*x+*y+*z;
}
int main()
{
    int x = 10, y=3;
    func(&y, &x, &x);
    printf("%d %d",x,y );
    return 0;
}
```

The value printed by above program is_

(A) 10, 3

(B) 31, 3

(C) 27, 7

(D) 27, 3



Array

Pointer arithmetic
Array definition
Declaration of array
[ ] operator
Initialization of array
Concept of base address and
Pointer
Memory diagram of array
Array index out of bound
Data type associated with array
Const association
Pointer arithmetic
Access element of array using
Index
Question



Pointer Arithmetic

- Multiplication & divison Not applied on Address

Addition & substraction with constant

Increment < post, pre

Decrement < post, pre

Two address (pointer) can be substracted

Slide



Pointer Arithmetic

```
int * ptr;
int a = 10;
ptr = &a;
```



int
4Byte

```
printf("%u", ptr+4);
```

↓

$\hookrightarrow ptr + 4 \times 4$

$= 100 + 4 \times 4 = 116$

$ptr + C = ptr + C \times$ Size of data type

C ↑ constant

Slide



Pointer Arithmetic

char = 1 Byte

```
char *ptr;
char ch = 'a';
ptr = &ch;
```

```
printf("%u", ptr+4);
```

ptr 100

ptr + 4 x 1

```
double *ptr;
double pi = 3.14;
ptr = &pi;
```

```
printf("%u", ptr+4);
```

ptr = ptr + 4 x 8

double 8 Byte

Slide



Pointer Arithmetic

ptr - c = ptr - c x Sizeof data type



Array

Array is collection of Similar Data type stored in Consecutive memory location

```
int a[4] = { 1,2,3,4 };
```

[ ] ← operator

array subscript operator

array Index operator

Slide

Array



1B ← 100 104 108 112 ← Consecutive Memory Locations

1B ← 101

1B ← 102

1B ← 103

Slide



Array

How to access array element



2 mins Summary

Topic — pointer Arithmetic

Topic — Array started

Topic

Topic

Topic

Slide

THANK - YOU

